Analysis

- *Stability of IQL* -

Kostrikov et al.

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1 Problem setup

Let $(\mathcal{A}, \mathcal{F}, \mu)$ be a probability space and let $\tau \in [0, 1)$. We consider the critic at a fixed state s, and write

$$q(a) \coloneqq Q(s, a), \qquad q \in L^\infty(\mathcal{A}, \mu).$$

Thus, throughout the local analysis below, q denotes the action-value fuction at a fixed state, viewed as an essentially bounded random variable over actions $a \sim \mu$. Let $\epsilon \in L^\infty(\mathcal{A}, \mu)$ denote a critic perturbation, and define the perturbed critic by

$$\hat{q} \coloneqq q + \epsilon.$$

Assume further that the relevant distributions are atomless. Define the ***expectile score function***

$$\psi_\tau(u) \coloneqq \tau u \mathbf{1}\{u \geq 0\} + (1 - \tau) u \mathbf{1}\{u < 0\},$$

and the corresponding ***expectile weight function***

$$w_\tau(u) \coloneqq \tau \mathbf{1}\{u \geq 0\} + (1 - \tau) \mathbf{1}\{u < 0\}.$$

Let

$$v_\tau \coloneqq e_\tau(q), \qquad \hat{v}_\tau \coloneqq e_\tau(q+\epsilon)$$

denote the respective τ-expectiles of the ideal and perturbed critics. The resulting expectile displacement is

$$\Delta_\tau \coloneqq \hat{v}_\tau - v_\tau. \tag{1}$$

Define its advantage $A_\tau \coloneqq q - v_\tau$. By translation invariance of the expectile operator, $e_\tau(A_\tau) = 0$, and

$$e_\tau(A_\tau + \epsilon) = e_\tau(q+\epsilon) - v_\tau = \Delta_\tau.$$

Hence, the expectile error induced by the critic perturbation can be written as

$$|\hat{v}_\tau - v_\tau| = |\Delta_\tau| = |e_\tau(A_\tau + \epsilon)|. \tag{2}$$

2 IQL Stability under Local Perturbations

For a fixed τ, consider the expectile operator

$$e_\tau : L^\infty(\mathcal{A}, \mu) \longrightarrow \mathbb{R}, \qquad q \longmapsto e_\tau(q).$$

The expectile $v_\tau = e_\tau(q)$ is characterized by the first-order estimating equation $G_{q,v_\tau} = 0$, where $G_\tau : L^\infty(\mathcal{A}, \mu) \times \mathbb{R} \longrightarrow \mathbb{R}$ is defined by

$$G_\tau(q, v) \coloneqq \mathbb{E}[\psi_\tau(q - v)]. \tag{3}$$

This implicit representation separates two directions of local variation: perturbations of the critic q and perturbations of the scalar expectile solution v. The corresponding differential structure provides the basis for distinguishing critic perturbations from residual errors in the expectile estimating equation.

Lemma 2.1 (Local Perturbations of the Expectile Solution). At (q, v_τ), the map G_τ is Fréchet differentiable with respect to q and differentiable with respect to v, with

$$D_q G_\tau(q, v_\tau)[\epsilon] = \mathbb{E}[w_\tau(A_\tau)\epsilon], \tag{4}$$

and

$$D_v G_\tau(q, v_\tau)[c] = -c\kappa_\tau, \qquad c \in \mathbb{R}. \tag{5}$$

where

$$\kappa_\tau \coloneqq \mathbb{E}[w_\tau(A_\tau)] = (1-\tau) + (2\tau - 1)\mu\{A_\tau > 0\}. \tag{6}$$

Since $\kappa_\tau \geq 1 - \tau > 0$, the implicit function theorem implies that the expectile operator e_τ is Fréchet differentiable at q with

$$De_\tau(q)[\epsilon] = \frac{\mathbb{E}[w_\tau(A_\tau)\epsilon]}{\kappa_\tau}. \tag{7}$$

Consequently, the expectile displacement Δ_τ defined in (1) admits the local expansion

$$\Delta_\tau = De_\tau(q)[\epsilon] + o(\|\epsilon\|_\infty). \tag{8}$$

Proof. See Appendix A. □

Critic perturbations

The derivative in (7) also shows that the expectile operator is locally non-expansive with respect to critic perturbations. Indeed, for any $\epsilon \in L^\infty(\mathcal{A}, \mu)$,

$$|De_\tau(q)[\epsilon]| = \left|\frac{\mathbb{E}[w_\tau(A_\tau)\epsilon]}{\kappa_\tau}\right| \leq \frac{\mathbb{E}[w_\tau(A_\tau)|\epsilon|]}{\kappa_\tau} \leq \|\epsilon\|_\infty,$$

where $\kappa_\tau = \mathbb{E}[w_\tau(A_\tau)]$. Hence,

$$\|De_\tau(q)\| \coloneqq \sup_{\substack{\epsilon \in L^\infty(\mathcal{A},\mu) \\ \epsilon \neq 0}} \frac{|De_\tau(q)[\epsilon]|}{\|\epsilon\|_\infty} \leq 1.$$

Moreover, equality is attained by the constant perturbation $\epsilon = 1$, for which

$$De_\tau(q)[1] = \frac{\mathbb{E}[w_\tau(A_\tau)]}{\kappa_\tau} = 1.$$

Therefore,

$$\|De_\tau(q)\| = 1. \tag{9}$$

Thus, though the local response in (7) contains the factor $1/\kappa_\tau$, critic perturbations are not amplified in the L^∞ norm. In particular, the vanishing of κ_τ alone does not imply instability with respect to critic perturbations.

Residual perturbations

The preceding analysis concerns perturbations of the critic itself. We now consider a distinct source of error arising from an inexact solution of the expectile estimating equation for a given critic. This error is not a critic perturbation; rather, it represents residual inaccuracies in expectile estimation, such as those arising from function approximation, optimization, or stochastic training.

Let $\tilde{v}_\tau$ denote an approximate expectile solution satisfying

$$G_\tau(q, \tilde{v}_\tau) + \zeta_\tau = 0,$$

where $\zeta_\tau \in \mathbb{R}$ denotes an additive residual in the expectile estimating equation. In contrast, the exact expectile satisfies $G_\tau(q, v_\tau) = 0$. The following proposition shows that local response to this residual is governed directly be the slope magnitude κ_τ.

Proposition 2.2 (Sensitivity to residual perturbations). Define the local conditioning co-

efficient of the τ-expectile equation by

$$\chi_\tau := \frac{1}{\kappa_\tau}. \tag{10}$$

For fixed τ, as $\zeta_\tau \to 0$, the displacement of the approximate expectile solution satisfies

$$\tilde{v}_\tau - v_\tau = \chi_\tau \zeta_\tau + o(\chi_\tau |\zeta_\tau|) = \frac{\zeta_\tau}{\kappa_\tau} + o\left(\frac{|\zeta_\tau|}{\kappa_\tau}\right). \tag{11}$$

Thus, unlike critic perturbations, residual errors in the expectile estimating equation are locally amplified by the inverse slope $1/\kappa_\tau$. Hence, χ_τ quantifies the local conditioning of the expectile solution with respect to residual perturbations.

Proof. See Appendix A. □

By Proposition 2.2, the behavior of κ_τ is important factor as the expectile level approaches its upper extreme. Let

$$q^* := \operatorname{esssup} q < \infty.$$

Recall that $\kappa = (1-\tau) + (2\tau - 1)\mu\{q > v_\tau\}$. As $\tau \uparrow 1$, the expectile approaches the essential supremum, $v_\tau \uparrow q^*$. Under the atomless assumption, continuity of probability from above gives $\mu\{q > v_\tau\} \to 0$. Since $1 - \tau \to 0$ and $2\tau - 1 \to 1$, it follows that

$$\kappa \to 0, \qquad \chi_\tau = \frac{1}{\kappa_\tau} \to \infty \qquad \text{as} \qquad \tau \uparrow 1. \tag{12}$$

Thus, the expectile equation becomes increasingly ill-conditioned with respect to residual perturbations in the extreme-expectile limit.

The vanishing of κ_τ also admits a natural local geometric interpretation through the expectile regression objective. Recall the asymmetric squared loss used for expectile regression in IQL, $L_\tau^2(u) := |\tau - \mathbf{1}\{u < 0\}|u^2$, and define

$$J_\tau(v) := \mathbb{E}[L_\tau^2(q - v)].$$

The expectile v_τ is the minimizer of J_τ, and its first-order condition coincides with the estimating equation $J_\tau'(v) = -2G_\tau(q, v)$. Moreover, the local curvature of the objective at v_τ is

$$J_\tau''(v_\tau) = 2\mathbb{E}[w_\tau(A_\tau)] = 2\kappa_\tau. \tag{13}$$

Therefore, $J_\tau''(v_\tau) \to 0$ as $\tau \uparrow 1$. The expectile objective therefore becomes increasingly flat around its minimizer as τ approaches one. In this sense, the vanishing of κ_τ can be viewed as a loss of local curvature: the same residual in the estimating equation can induce a larger displacement of the expectile solution.

3 Error through Bellman Iteration

The preceding analysis was carried out at a fixed state s_0, where $q(a) = Q(s_0, a)$ and $v_\tau = e_\tau(q)$. We now extend these fixed-state quantities over the state space and introduce the Bellman iteration index k. Specifically, collecting the action-value functions $q(a) = Q_k(s_0, a)$ over all states gives the critic $Q_k(s, a)$, while collecting the corresponding expectiles v_τ gives the state-value function

$$V_{k,\tau} \coloneqq e_\tau(Q_k(s, \cdot)).$$

Accordingly, the centered action-value function is defined by $A_{k,\tau}(s, a) \coloneqq Q_k(s, a) - V_{k,\tau}(s)$, and the local slope of the expectile estimating equation becomes

$$\kappa_{k,\tau}(s) \coloneqq \mathbb{E}_{a\sim\mu(\cdot|s)}[w_\tau(A_{k,\tau}(s, a))].$$

Likewise, the fixed-state critic perturbation ϵ extends to an iteration-dependent perturbation $\epsilon_k(s, a)$,

$$\hat{Q}_k(s, a) = Q_k(s, a) + \epsilon_k(s, a).$$

Moreover the residual perturbation ζ_τ becomes $\zeta_{k,\tau}(s)$.

Our objective is to lift the preceding one-step perturbation analysis to the Bellman iteration. We examine how this error enters subsequent critic updates through the Bellman equation. This yields a recursive description of thow the two error channels propagate and accumulate across iterations.

We now extencd the preceding fixed-state perturbation analysis to the state-vvalue function. Define the total state-value disturbance at iteration k by

$$g_k(s) \coloneqq \tilde{V}_{k,\tau}(s) - V_{k,\tau}(s).$$

At each state s, this disturbance can be decomposed as

$$g_k(s) = (\hat{V}_{k,\tau}(s) - V_{k,\tau}(s)) + (\tilde{V}_{k,\tau}(s) - \hat{V}_{k,\tau}(s))$$

where $\hat{V}_{k,\tau}(s) - V_{k,\tau}(s)$ and $\tilde{V}_{k,\tau}(s) - \hat{V}_{k,\tau}(s)$ means critic perturbation and residual perturbation, respectively. Applying (8) and residual perturbation by (11) gives

$$g_k(s) = De_\tau(Q_k(s, \cdot))[\epsilon_k(s, \cdot)] + \frac{\zeta_{k,\tau}(s)}{\kappa_{k,\tau}(s)} + r_{k,\tau}(s), \tag{14}$$

where, pointwise in s,

$$r_{k,\tau}(s) = o(\|\epsilon_k(s, \cdot)\|_\infty) + o\left(\frac{|\zeta_{k,\tau}(s)|}{\kappa_{k,\tau}(s)}\right). \tag{15}$$

Thus, the one-step state-value disturbance retains the two channels identified in the fixed-state analysis: the critic perturbation enters through the derivative of the expectile opera-

tor, whereas the residual perturbation enters through the inverse local slope $1/\kappa_{k,\tau}(s)$.

We now propagate g_k through the Bellman update. Let B_k denote the bounded linear operator describing the local response of the critic-fitting procedure to an error in its Bellman target at iteration k. Specifically, if h denotes an error in the Bellman target, $B_k h$ represents the corresponding first-order error in the fittted critic. Since the state-value disturbance g_k induces an error γg_k in the Bellman target, the next critic perturbation satisfies

$$\epsilon_{k+1} = \gamma B_k g_k + e_{k+1} + \eta_{k+1}, \tag{16}$$

where e_{k+1} denotes an additional systematic critic fitting error and η_{k+1} a new stochastic error satisfying $\mathbb{E}[\eta_{k+1}|\mathcal{F}_k] = 0$. Substituting (14) into (16) yields

$$\epsilon_{k+1} = M_{k,\tau}\epsilon_k + N_{k,\tau}\zeta_{k,\tau} + e_{k+1} + \eta_{k+1} + \rho_{k,\tau}, \tag{17}$$

where

$$M_{k,\tau} \coloneqq \gamma B_k D e_\tau(Q_k), \qquad N_{k,\tau}\zeta \coloneqq \gamma B_k \left(\frac{\zeta}{\kappa_{k,\tau}}\right) \qquad \text{and} \qquad \rho_{k,\tau} \coloneqq \gamma B_k r_{k,\tau}.$$

This recursion separates the two mechanisms through which errors enter Bellman iteration. The term $M_{k,\tau}\epsilon_k$ propagates the existing critic perturbation, whereas $N_{k,\tau}\zeta_{k,\tau}$ represents the critic error induced by the residual perturbation at the current expectile update. The remaining terms account for newly introduced critic-fitting errors and propagated higher-order effects.

To isolate the state-wise scaling appearing in the residual propagation operator $N_{k,\tau}$, define the multiplication operator $K_{k,\tau} : L^\infty(\mathcal{S}) \to L^\infty(\mathcal{S})$ by

$$[K_{k,\tau} z](s) \coloneqq \frac{z(s)}{\kappa_{k,\tau}(s)}. \tag{18}$$

Then, $N_{k,\tau} = \gamma B_k K_{k,\tau}$.

Lemma 3.1 Suppose that $\inf_s \kappa_{k,\tau}(s) > 0$. Then, $K_{k,\tau}$ is a bounded linear operator on the space of bounded state functions with $\|K_{k,\tau}\| = 1/\inf_s \kappa_{k,\tau}(s)$.

Proof. For any bounded state function z,

$$\|K_{k,\tau} z\|_\infty = \sup_s \frac{|z(s)|}{\kappa_{k,\tau}(s)} \le \frac{\|z\|_\infty}{\inf_s \kappa_{k,\tau}(s)}.$$

Hence,

$$\|K_{k,\tau}\| \le \frac{1}{\inf_s \kappa_{k,\tau}(s)}.$$

Converselu, choosing bounded functions concentrated arbitrarily near states at which $\kappa_{k,\tau}$ approaches its infimum gives the reverse inequality. Therefore,

$$\|K_{k,\tau}\| = \frac{1}{\inf_s \kappa_{k,\tau}(s)}.$$

□

Proposition 3.2 *(Norm bounds for the two propagation channels).* Consider the bounded linear operators

$$M_{k,\tau} : L^\infty(\mathcal{S} \times \mathcal{A}) \to L^\infty(\mathcal{S} \times \mathcal{A}), \qquad M_{k,\tau} = \gamma B_k De_\tau(Q_k),$$

and

$$N_{k,\tau} : L^\infty(\mathcal{S}) \to L^\infty(\mathcal{S} \times \mathcal{A}), \qquad N_{k,\tau} = \gamma B_k K_{k,\tau}.$$

Then,

$$\|M_{k,\tau}\| \le \gamma \|B_k\|, \tag{19}$$

while provided $\inf_s \kappa_{k,\tau}(s) > 0$,

$$\|N_{k,\tau}\| \le \frac{\gamma \|B_k\|}{\inf_s \kappa_{k,\tau}(s)}. \tag{20}$$

Thus, the critic-perturbation channel is controlled independently of the expectile conditioning coefficient, whereas the worst-case bound for the residual channel scales with $1/\inf_s \kappa_{k,\tau}(s)$.

Proof. By the fixed-state result (9), $\|De_\tau(q)\| = 1$. Applying this result state-wise gives $\|De_\tau(Q_k)\| = 1$. Therefore,

$$\|M_{k,\tau}\| \le \gamma \|B_k\| \|De_\tau(Q_k)\| = \gamma \|B_k\|.$$

Similarly, Lemma 3.1 yields

$$\|N_{k,\tau}\| \le \gamma \|B_k\| \|K_{k,\tau}\| = \frac{\gamma \|B_k\|}{\inf_s \kappa_{k,\tau}(s)}.$$

□

4 Local Stability

We now examine how the preceding local error recursion behaves over repeated Bellman iterations. Define

$$d_{k+1} \coloneqq N_{k,\tau}\zeta_{k,\tau} + e_{k+1} + \eta_{k+1} + \rho_{k,\tau}, \tag{21}$$

so that $\epsilon_{k+1} = M_{k,\tau}\epsilon_k + d_{k+1}$. For any $\alpha > 0$, Young's inequality with $p = q = 2$ gives

$$\mathbb{E}\|\epsilon_{k+1}\|_\infty^2 \le (1+\alpha)\mathbb{E}[\|M_{k,\tau}\|^2 \|\epsilon_k\|_\infty^2] + \left(1 + \frac{1}{\alpha}\right) \mathbb{E}\|d_{k+1}\|_\infty^2. \tag{22}$$

Suppose that $\|M_{k,\tau}\| \le m < 1$ almost surely and uniformly in k. Choose $\alpha > 0$ such that $(1+\alpha)m^2 < 1$, and define $a \coloneqq (1+\alpha)m^2$. Then,

$$\mathbb{E}\|\epsilon_{k+1}\|_\infty^2 \le a\mathbb{E}[\|\epsilon_k\|_\infty^2] + \left(1 + \frac{1}{\alpha}\right) \mathbb{E}\|d_{k+1}\|_\infty^2. \tag{23}$$

Iterating this inequality gives, for $K \geq 1$,

$$\mathbb{E}\|\epsilon_K\|_\infty^2 \leq a^K \mathbb{E}\|\epsilon_0\|_\infty^2 + \left(1 + \frac{1}{\alpha}\right) \sum_{j=0}^{K-1} a^{K-1-j} \mathbb{E}\|d_{j+1}\|_\infty^2.$$

In particular, if $\sup_k \mathbb{E}\|d_{k+1}\|_\infty^2 \leq D < \infty$, then

$$\limsup_{K\to\infty} \mathbb{E}\|\epsilon_K\|_\infty^2 \leq \frac{(1+\alpha^{-1})D}{1-a}.$$

Hence, though new forcing errors are introduced at every Bellman iteration, their past effects are geometrically attenuated when the critic-propagation channel is contractive. As a result, uniformly bounded forcing errors yield a uniformly bounded critic error. Morover, if $\mathbb{E}\|d_k\|_\infty^2 \to 0$, then $\mathbb{E}\|\epsilon_k\|_\infty^2 \to 0$, and hence $\epsilon_k \to 0$ in probability by Markov's inequality.

The preceding perturbation analysis is local at each fixed state. To extend this analysis uniformly over a relevant state domain, we require a mild regularity condition on the conditional distribution of the action values near the state-dependent expectile. The following proposition shows that joint continuity of the conditional CDF is sufficient for this purpose. This condition does not require absolute continuity or the existence of a conditional density, and thus still allows singular continuous distributions; it only excludes atomic components. Moreover, the same condition implies the conditional atomlessness used in the preceding fixed-state analysis.

Proposition 4.1 Let $\mathcal{S}_0 \subset \mathcal{S}$ be compact, and fix an iteration k. Suppose that

$$F_k(s,x) \coloneqq \mu_s\{Q_k(s,\mathcal{A}) \leq x\}$$

is jointly continuous in (s,x) on $\mathcal{S}_0 \times I_k$, where $I_k \subset \mathbb{R}$ is a compact interval containing a neighborhood of the essential range of Q_k over $\mathcal{S}_0$. Then,

$$\omega_{k,\tau}(\xi) \coloneqq \sup_{s\in\mathcal{S}_0} \mu_s\{|A_{k,\tau}(s,\mathcal{A})| \leq \xi\} \longrightarrow 0 \qquad \text{as} \qquad \xi \downarrow 0. \tag{24}$$

Moreover, for every $s \in \mathcal{S}_0$, the conditional distribution of $Q_k(s,\mathcal{A})$ is atomless.

Proof. See Appendix A. □

(24) allows the preceding pointwise local expansion to be controlled uniformly over the relevant state domain. In particular, the nonlinear remainder becomes negligible relative to the first-order perturbations as the local radius shrinks.

Lemma 4.2 Fix an iteration k, and suppose that (24) holds. Then, there exist non-negative functions $c_{k,\tau}(\delta)$ and $h_{k,\tau}(\delta)$ such that

$$c_{k,\tau}(\delta) \to 0, \qquad h_{k,\tau}(\delta) \to 0, \qquad \text{as} \qquad \delta \downarrow 0,$$

and, whenever $\|\epsilon_k\|_\infty + \|K_{k,\tau}\zeta_{k,\tau}\|_\infty \le \delta$, the local remainder satisfies

$$\|r_{k,\tau}\|_\infty \le c_{k,\tau}(\delta)\|\epsilon_k\|_\infty + h_{k,\tau}(\delta)\|K_{k,\tau}\zeta_{k,\tau}\|_\infty. \tag{25}$$

Consequently,

$$\|\rho_{k,\tau}\|_\infty \le \gamma\|B_k\|[c_{k,\tau}(\delta)\|\epsilon_k\|_\infty + h_{k,\tau}(\delta)\|K_{k,\tau}\zeta_{k,\tau}\|_\infty.] \tag{26}$$

Proof. See Appendix A. □

Theorem 4.3 *(Finite-Horizon Local Stability).* Consider the local critic-error recursion

$$\epsilon_{k+1} = M_{k,\tau}\epsilon_k + N_{k,\tau}\zeta_{k,\tau} + f_{k+1} + \rho_{k,\tau}, \qquad 0 \le k < T, \tag{27}$$

where $f_{k+1} \coloneqq e_{k+1} + \eta_{k+1}$. Suppose that the conditions of Proposition 4.1 hold for each $k < T$, and that

$$\underline{\kappa}_T \coloneqq \min_{0\le k<T} \inf_{s\in\mathcal{S}_0} \kappa_{k,\tau}(s) > 0, \qquad \sup_{0\le k<T} \|\zeta_{k,\tau}\|_\infty < \infty.$$

Assume further that the aggregate fitting error satisfies

$$P\left(\|f_{k+1}\|_\infty > x \,|\, \mathcal{F}_k\right) \le q_k(x) \qquad \text{a.s.} \tag{28}$$

and that the critic propagation and forcing terms are sufficiently small to perserve a local radius $\delta > 0$. Then, for every $\beta \in (0,1)$, provided $\|\epsilon_0\|_\infty \le \delta$,

$$P\left(\max_{0\le k\le T}\|\epsilon_k\|_\infty \le \delta\right) \ge 1 - \beta. \tag{29}$$

Proof. see Appendix A. □

A Proofs

Proof of Lemma 2.1. Since $G_\tau(q, v_\tau) = 0$, consider a perturbation $\epsilon \in L^\infty(\mathcal{A}, \mu)$ of q. For every a such that $A_\tau \neq 0$, the score function ψ_τ is lineqr in a neighborhood of $A_\tau(a)$, and hence

$$\frac{d}{dt}\psi_\tau(A_\tau + t\epsilon(a))\Big|_{t=0} = w_\tau(A_\tau(a))\epsilon(a).$$

Since $\mu\{A_\tau = 0\} = 0$ and ψ_τ is globally Lipschitz, dominated convergence theorem yields

$$D_q G_\tau(q, v_\tau)[\epsilon] = \lim_{t\to 0} \frac{G_\tau(q + t\epsilon, v_\tau) - G_\tau(q, v_\tau)}{G_\tau(q, v_\tau)} = \mathbb{E}[w_\tau(A_\tau)\epsilon].$$

Similarly, for $c \in \mathbb{R}$,

$$D_v G_\tau(q, v_\tau)[c] = -c\mathbb{E}[w_\tau(A_\tau)] = -c\kappa.$$

Since $\kappa = \mathbb{E}[w_\tau(A_\tau)] \geq \min\{\tau, 1 - \tau\} > 0$, $D_v G_\tau(q, v_\tau)$ is invertible. Thus, the implicit function theorem gives

$$De_\tau(q)[\epsilon] = -((D_v G_\tau(q, v_\tau))^{-1} \circ D_q G_\tau(q, v_\tau)[\epsilon]) = \frac{\mathbb{E}[w_\tau(A_\tau)\epsilon]}{\kappa_\tau}.$$

It remians to verify that the derivative with respect to q is Fréchet. Define

$$R_\epsilon \coloneqq \psi_\tau(A_\tau + \epsilon) - \psi_\tau(A_\tau) - w_\tau(A_\tau)\epsilon.$$

Since ψ_τ is linear on each side of the origin,

$$R_\epsilon(a) = 0 \qquad \text{whenever} \qquad A_\tau(a)(A_\tau(a) + \epsilon(a)) > 0.$$

Thus, a nonzero remainder can occur only when the perturbation crosses the origin, which implies

$$|A_\tau(a)| \leq |\epsilon(a)| \leq \|\epsilon\|_\infty.$$

Since ψ_τ is globally Lipshcitz, for some finite constant C_τ,

$$|R_\epsilon(a)| \leq C_\tau \|\epsilon\|_\infty \mathbf{1}\{|A_\tau| \leq \|\epsilon\|_\infty\}.$$

Consequently,

$$\frac{\mathbb{E}|R_\epsilon|}{\|\epsilon\|_\infty} \leq C_\tau \mu\{|A_\tau| \leq \|\epsilon\|_\infty\}.$$

SInce $\mu\{A_\tau = 0\} = 0$, continuity from above gives $\mu\{|A_\tau| \leq \delta\} \to 0$ as $\delta \downarrow 0$. Hence, $\mathbb{E}|R_\epsilon| = o(\|\epsilon\|_\infty)$, which establishes the required Fréchet differentiability and completes the proof.

□

Proof of Proposition 2.2. Let $\Delta_\tau^{\text{res}} \coloneqq \tilde{v}_\tau - v_\tau$. Since $G_\tau(q, v_\tau) = 0$ and $G_\tau(q, \tilde{v}_\tau) + \zeta_\tau = 0$, a

first-order expansion with respect to v around v_τ gives

$$-\zeta_\tau = D_v G_\tau(q, v_\tau)[\Delta_\tau^{\mathrm{res}}] + o(|\Delta_\tau^{\mathrm{res}}|).$$

By Lemma 2.1, $D_v G_\tau(q, v_\tau)[c] = -\kappa_\tau c$, and hence

$$\zeta_\tau = \kappa_\tau \Delta_\tau^{\mathrm{res}} + o(|\Delta_\tau^{\mathrm{res}}|).$$

Since $\kappa_\tau > 0$ is fixed, this implies

$$\Delta_\tau^{\mathrm{res}} = O\left(\frac{|\zeta_\tau|}{\kappa_\tau}\right) \qquad \text{as} \qquad \zeta_\tau \to 0.$$

Substituting this into the remainder term yields

$$\Delta_\tau^{\mathrm{res}} = \frac{\zeta_\tau}{\kappa_\tau} + o\left(\frac{|\zeta_\tau|}{\kappa_\tau}\right) \qquad \text{as} \qquad \zeta_\tau \to 0.$$

□

Proof of Proposition 4.1. Since $\mathcal{S}_0 \times I_k$ is compact and F_k is jointly continuous, F_k is uniformly continuous. In particular, for every fixed $s \in \mathcal{S}_0$, the map $x \mapsto F_k(s, x)$ is continuous. Hence,

$$\mu_s(Q_k(s, \mathcal{A}) = x) = F_k(s, x) - F_k(s, x-) = 0$$

for every $x \in I_k$, so the conditional distribution of $Q_k(s, \mathcal{A})$ is atomless.
Define the uniform modulus

$$\Omega_k(\xi) := \sup_{\substack{s \in \mathcal{S}_0 \\ x, y \in I_k, |x-y| \le \xi}} |F_k(s, x) - F_k(s, y)|.$$

Uniform continuity implies $\Omega_k(\xi) \to 0$ as $\xi \downarrow 0$. Since $V_{k,\tau}(s)$ lies in the essential range interval of $Q_k(s, \cdot)$, for all sufficiently small $\xi > 0$,

$$\begin{aligned}
\mu_s\{|A_{k,\tau}(s, \mathcal{A})| \le \xi\} &= \mu_s\{|Q_k(s, \mathcal{A}) - V_{k,\tau}(s)| \le \xi\} \\
&= F_k(s, V_{k,\tau}(s) + \xi) - F_k(s, V_{k,\tau}(s) - \xi) \\
&\le \Omega_k(2\xi).
\end{aligned}$$

Therefore, the assertion holds. □

Proof of Lemma 4.2. The expectile score function ψ_τ is piecewise linear, with its slope changing only at the origin. Hence, if the perturbation preserves the sign of $A_{k,\tau}(s, a)$, the perturbed and unperturbed advantages remain on the same linear branch of ψ_τ, and the first-order expansion is exact. Thus, $r_{k,\tau}(s) = 0$ in the absence of boundary crossings. Under the local condition

$$\|\epsilon_k\|_\infty + \|K_{k,\tau}\zeta_{k,\tau}\|_\infty \le \delta,$$

the induced displacement of the advantage is bounded by $C\delta$ for some finite local con-

stant C. Consequently, a boundary crossing can occur only if

$$|A_{k,\tau}(s,a)| \le C\delta.$$

Since ψ_τ is globally Lipschitz, the pointwise remainder on this crossing region is bounded by a constant multiple of the perturbation magnitude. Hence, after integration with respect to μ_s, the two perturbation channels satisfy bounds of the form

$$\|r_{k,\tau}\|_\infty \le C_1\omega_{k,\tau}(C_2\delta)\|\epsilon_k\|_\infty + C_3\omega_{k,\tau}(C_4\delta)\|K_{k,\tau}\zeta_{k,\tau}\|_\infty,$$

for some finite constants C_1, C_2, C_3, C_4, where $\omega_{k,\tau}(\cdot)$ is the function defined in Proposition 4.1. Define

$$c_{k,\tau}(\delta) \coloneqq C_1\omega_{k,\tau}(C_2\delta), \qquad h_{k,\tau}(\delta) \coloneqq C_3\omega_{k,\tau}(C_4\delta).$$

Then, by (24), it follows that $c_{k,\tau}(\delta) \to 0$ and $h_{k,\tau}(\delta) \to 0$ as $\delta \downarrow 0$. Finally, using $\rho_{k,\tau} = \gamma B_k r_{k,\tau}$, the assertion holds.

□

Proof of Theorem 4.3. Since the horizon T is finite, define

$$b_T \coloneqq \max_{0\le k<T} \gamma\|B_k\|, \qquad \underline{\kappa}_T \coloneqq \min_{0\le k<T} \inf_{s\in\mathcal{S}_0} \kappa_{k,\tau}(s) > 0,$$

and

$$Z_T \coloneqq \max_{0\le k<T} \|\zeta_{k,\tau}\|_\infty.$$

The operator bounds obtained above give, for every $k < T$,

$$\|M_{k,\tau}\| \le b_T, \qquad \|K_{k,\tau}\zeta_{k,\tau}\|_\infty \le \frac{Z_T}{\underline{\kappa}_T}, \tag{30}$$

and consequently

$$\|N_{k,\tau}\zeta_{k,\tau}\|_\infty \le b_T\frac{Z_T}{\underline{\kappa}_T}. \tag{31}$$

For a given $\beta \in (0,1)$, choose $b_{T,\beta} > 0$ such that

$$\sum_{k=0}^{T-1} q_k(b_{T,\beta}) \le \beta,$$

and define the event

$$E_{T,\beta} \coloneqq \bigcap_{k=0}^{T-1} \{\|f_{k+1}\|_\infty \le b_{T,\beta}\}.$$

By the conditional tail bound, the tower property, and the union bound,

$$P(E_{T,\beta}^c) \le \sum_{k=0}^{T-1} P(\|f_{k+1}\|_\infty > b_{T,\beta}) \le \sum_{k=0}^{T-1} q_k(b_{T,\beta}) \le \beta.$$

Hence,

$$P(E_{T,\beta}) \geq 1 - \beta. \tag{32}$$

Now, suppose that $\|\epsilon_k\|_\infty \leq \delta$. From (30),

$$\|\epsilon_k\|_\infty + \|K_{k,\tau}\zeta_{k,\tau}\|_\infty \leq \delta + \frac{Z_T}{\underline{\kappa}_T} =: U_T.$$

By Lemma 4.2,

$$\|\rho_{k,\tau}\|_\infty \leq b_T c_T(U_T)\|\epsilon_k\|_\infty + b_T h_T(U_T)\frac{Z_T}{\underline{\kappa}_T}, \tag{33}$$

where

$$c_T(U_T) \coloneqq \max_{0\leq k<T} c_{k,\tau}(U_T), \qquad h_T(U_T) \coloneqq \max_{0\leq k<T} h_{k,\tau}(U_T),$$

□

Using the critic-error recursion together with (30) - (33), we obtain

$$\|\epsilon_{k+1}\|_\infty \leq b_T(1 + c_T(U_T))\|\epsilon_k\|_\infty + b_T(1 + h_T(U_T))\frac{Z_T}{\underline{\kappa}_T} + b_{T,\beta}. \tag{34}$$

Define $m_T \coloneqq b_T(1 + c_T(U_T))$. since $b_T < 1$ and $c_T(\xi) \to 0$ as $\xi \downarrow 0$, the local radius and residual level may be chosen sufficiently small so that $b_T < 1$. Under the assumption, we can set

$$b_T(1 + h_T(\Delta_T))\frac{Z_T}{\underline{\kappa}_T} + b_{T,\beta} \leq (1 - m_T)\delta, \tag{35}$$

(34) implies that whenever $\|\epsilon_k\|_\infty \leq \delta$, Hence, the ball $\{\epsilon \mid \|\epsilon\|_\infty \leq \delta\}$ is forward invariant on $E_{T,\beta}$. Since $\|\epsilon_0\|_\infty \leq \delta$, induction gives $\max_{0\leq k\leq T}\|\epsilon_k\|_\infty \leq \delta$ on $E_{T,\beta}$. Combining this with (32) yields

$$P\left(\max_{0\leq k\leq T}\|\epsilon_k\|_\infty \leq \delta\right) \geq 1 - \beta,$$

which proves the result.